



THE UNIVERSITY OF
MELBOURNE

ACCT90002

Financial Statement Analysis

Week 3: Accounting analysis (1) The principles



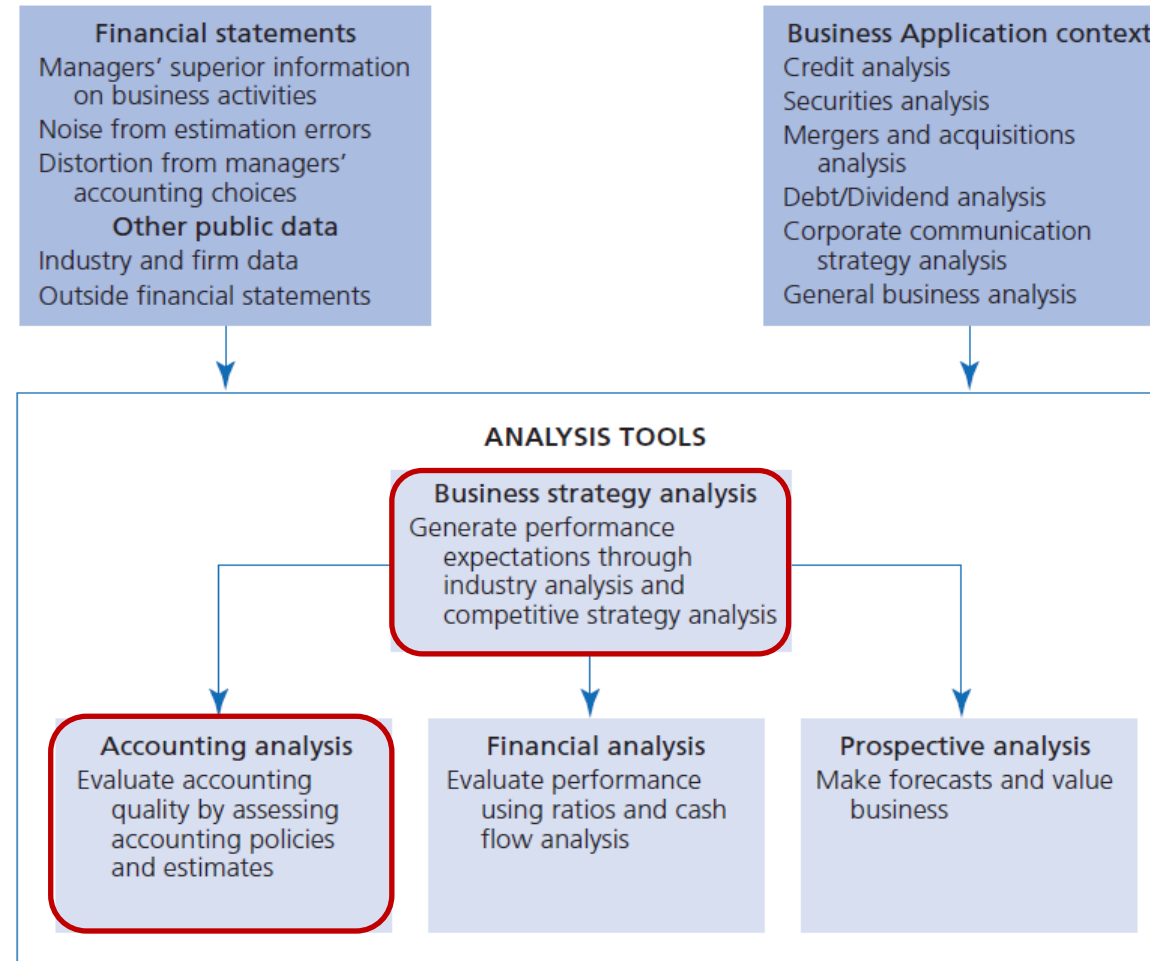


What we know so far about FSA...

- Why: Information asymmetry and agency problem
- What: FSA is the set of techniques that enable inside information to be incorporated in corporate valuations and therefore enable the efficient resource allocation – an art and a science
- How: FSA takes place within a structured framework, which start with Business Strategy Analysis

Framework - Financial Statement Analysis

FIGURE 1.3 Analysis using financial statements





Accounting analysis – [Weeks 3-4]

- It aims to evaluate the degree to which a firm's accounting captures the underlying business economics (week3)
- It enables analysts to assess the degree of distortion in a firm's accounting numbers (week3 & 4)
- It enables the analyst to recast accounting numbers to create unbiased accounting numbers (week4)
- Sound accounting analysis improves the reliability of conclusions from financial analysis



Accounting analysis – purpose

- There is an **economic reality** in which shareholders/creditors/users invest
- The shareholders/creditors/users cannot observe the **economic reality**, they can only observe the **financial reporting**
- There are **choices made in accounting** that mean the accounting may not fully reflect the **economic reality**
- Accounting analysis provides a means to better understand the **economic reality** from the financial reports

Accounting analysis-purpose

- Financial reporting information is **noisy and biased**
 - even in the presence of accounting **regulation** and external **auditing**
- If potential distortions are large, accounting analysis can add considerable value
 - It is an important *precondition* for effective financial analysis
 - The quality of **financial analysis** and the inferences drawn depend on the quality of the *underlying accounting information*, the *raw material* for analysis

"noisy" refers to random fluctuations
"biased" refers to systematic distortions

The rule of “Garbage in, garbage out” in computer science also applies in Financial Statements Analysis

Examples



China Huishan Dairy Holdings
Company Limited

MUDDY
WATERS
RESEARCH

CHINA HUIZHAN DAIRY HOLDINGS

6863 HK Two Days 0.42 -2.38 85.00%

Reports surfaced alleging funds
misappropriation and financial
statements manipulation.

Bloomberg



Investors wake up to Luckin's alleged fraud

Luckin Coffee share price in \$



Source: Bloomberg

<https://www.businessinsider.com/huishan-dairy-stock-crashes-hong-kong-minutes-2017-3>

<https://www.ft.com/content/2e0ef250-fa12-4183-93ff-e404f0cc0fc5>

Factors influencing the financial statements

Transactions, events and conditions

- Managers may use their accounting discretion to distort reported profits

Accounting standards: IFRS

- Adoption of IFRS reduces the cost of analysis and limits managers' ability to distort financial statements

Auditing

- Auditing improves the credibility of accounting data

Legal environment

- The threat of lawsuits and resulting penalties can improve the accuracy of disclosure

Governance

- Good corporate governance is designed to promote investor confidence

Influences on Accounting





Influences on Accounting

- Many of the influences on accounting result in “better” or less biased accounting
- **Incentives** can have effects that encourage managers to make accounting choices that result in earnings that may include elements of accounting bias
- When this occurs, we refer to it as **earnings management**, which is different to accounting fraud.
 - **Earnings management** choices are not “untrue” or “false”, but choices available to managers that, when collectively made, can influence the economic narrative gathered from the financial statements

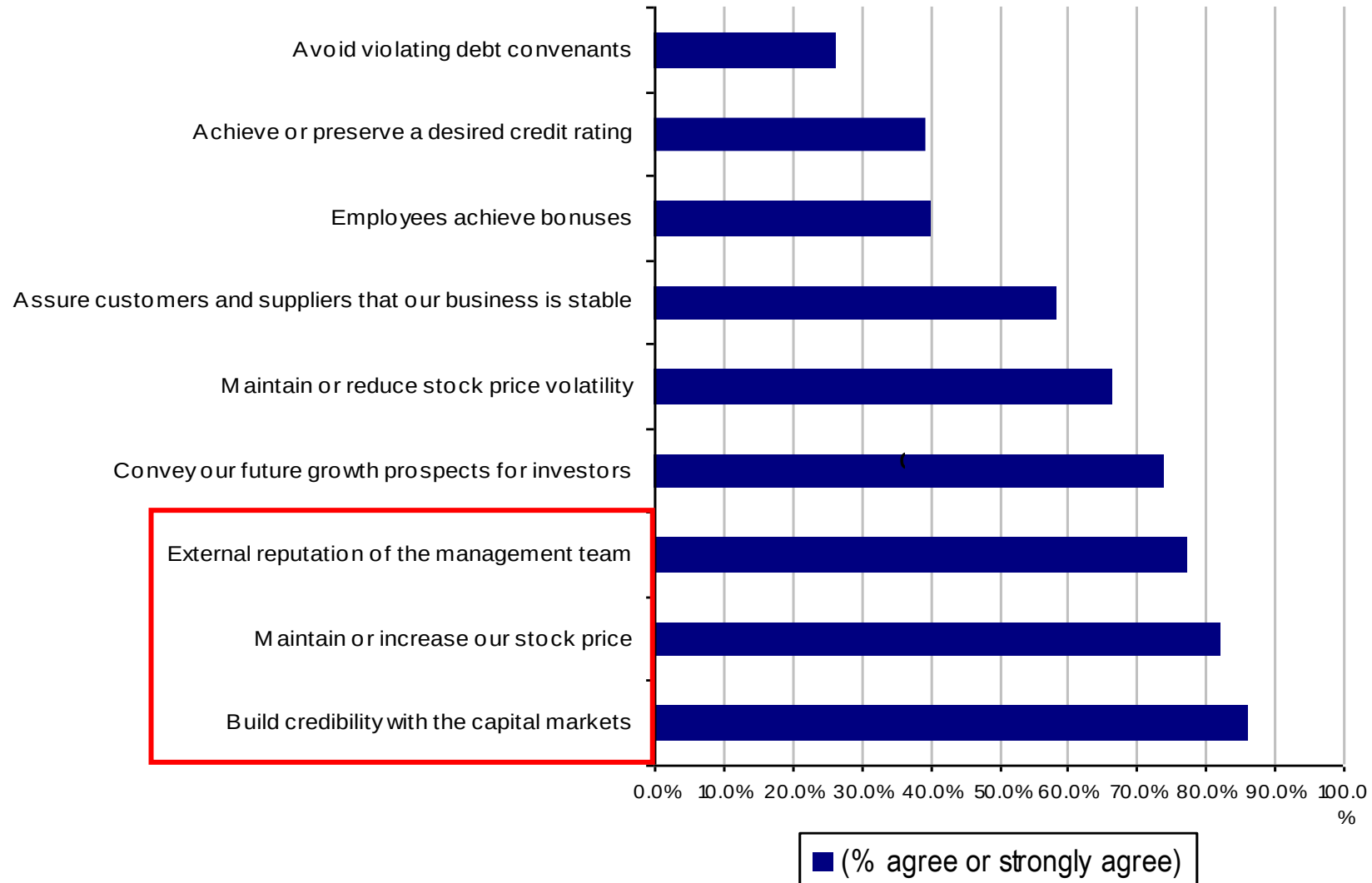
-



Motives to manage earnings

- Capital market (share prices, IPOs, takeovers, market expectation)
- Debt covenants (to avoid breaching debt covenants)
- Political motives (to avoid undesirable public attention/scrutiny)
- Executive compensation (to maximize current+ PV of future compensation)

“Meeting earnings benchmarks helps ...”



“Failing to meet benchmarks...”





Facebook (FB:NASDAQ)

Real Time Quote | Last Trade from NYSE Arca, Volume from CTA | USD

Extended Hours | 08/17/2018

173.30 -0.5 (-0.29%)

Volume

334,631

Close | 08/17/2018

173.80 -0.9 (-0.52%)

Volume

24,366,835

52 week range

149.02 - 218.62

1D

5D

1M

3M

6M

YTD

1Y

5Y

ALL

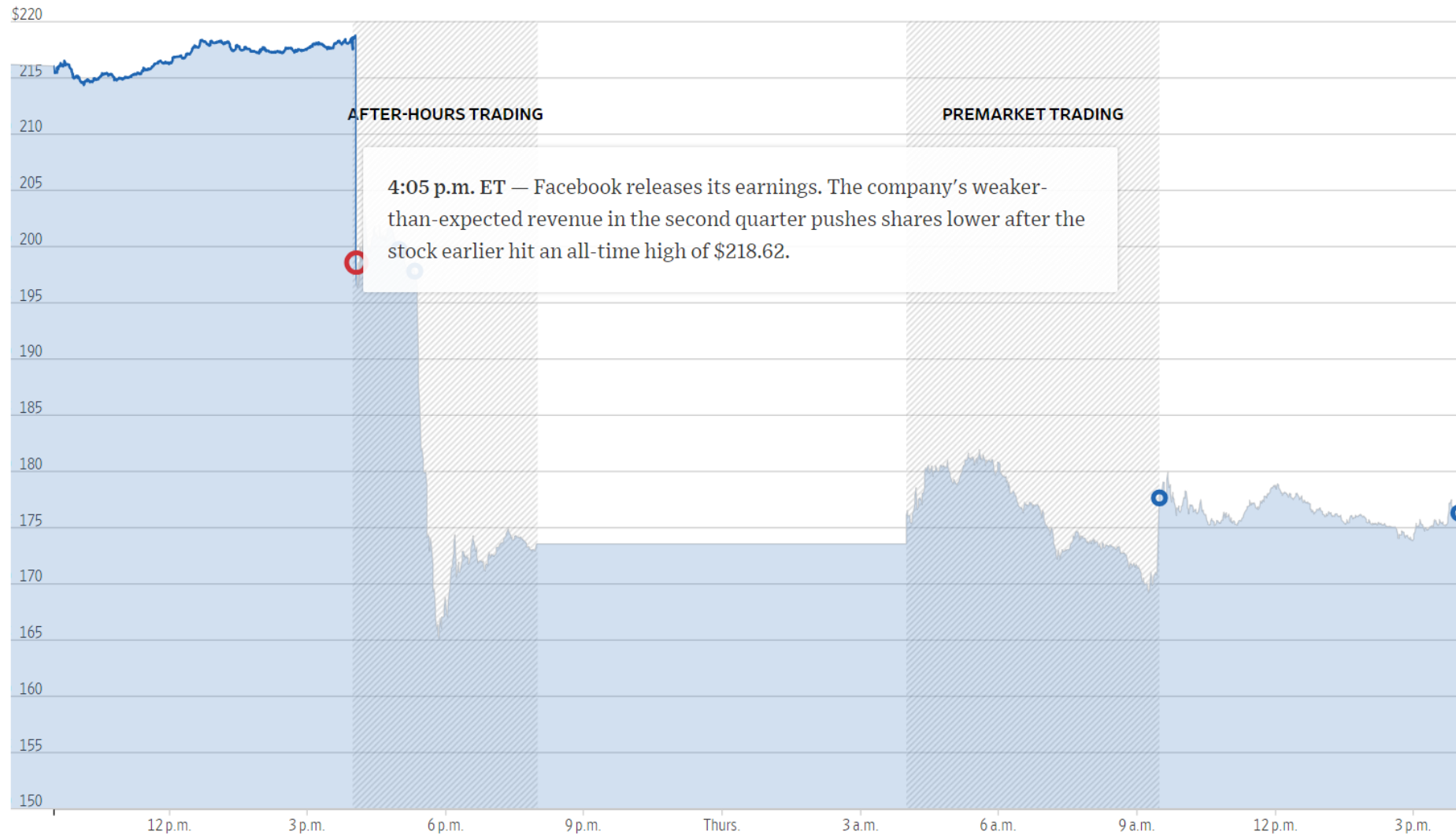
+ Comparison

1D

Display

Studies







<https://www.reuters.com/technology/intel-shares-set-fall-most-24-years-it-struggles-with-turnaround-2024-08-02/>

NasdaqGS - Nasdaq Real Time Price • USD

Intel Corporation (INTC)

☆ Follow

↕ Compare

21.48 **-7.57 (-26.06%)** **21.58** **+0.10 (+0.47%)**

At close: 2 August at 4:00 pm GMT-4

After hours: 2 August at 7:59 pm GMT-4



Stock market reacts to disappointing outlook

Intel shares plunged over 20% in pre-market trading on Thursday, following the company's announcement of drastic cost-cutting measures and disappointing earnings forecast. The steep decline reflects growing concerns about the chipmaker's ability to navigate challenges in the semiconductor industry and execute its turnaround plan.

Earnings fall short of expectations

For the second quarter (Q2), Intel reported revenue of \$12.8 billion, a 1% decrease year-over-year and below Wall Street's expectations of \$12.9 billion. The company's pro forma earnings came in at 2 cents per share, down from 13 cents the previous year and significantly lower than the 10 cents analysts had anticipated.

Weak forecast raises eyebrows

Intel's outlook for the third quarter (Q3) further dampened investor sentiment. The company projected revenue between \$12.5 billion and \$13.5 billion, with a potential pro forma loss of 3 cents per share. This forecast falls short of Wall Street's expectations of a 13 cent per share profit on revenue of approximately \$14.4 billion.

So, what kind of expectations are

- Remember, we are talking about expected future returns on capital made by the company
- With this in mind, investors have certain “**psychological**” benchmarks that add to comfort when investing
 - Meeting or beating expectations
 - Earnings/cash flows > 0
 - Earnings/cash flow growth > 0

Detect EM: The Burgstahler and Dichev (1997) graph

- **MEET EARNINGS EXPECTATIONS.** Significant negative effects on share price and manager reputation if expectations not met (prospect theory)

For example, earnings > 0 :

How likely is it that this pattern occurred at random?

- This question is the basis of much of forensic accounting

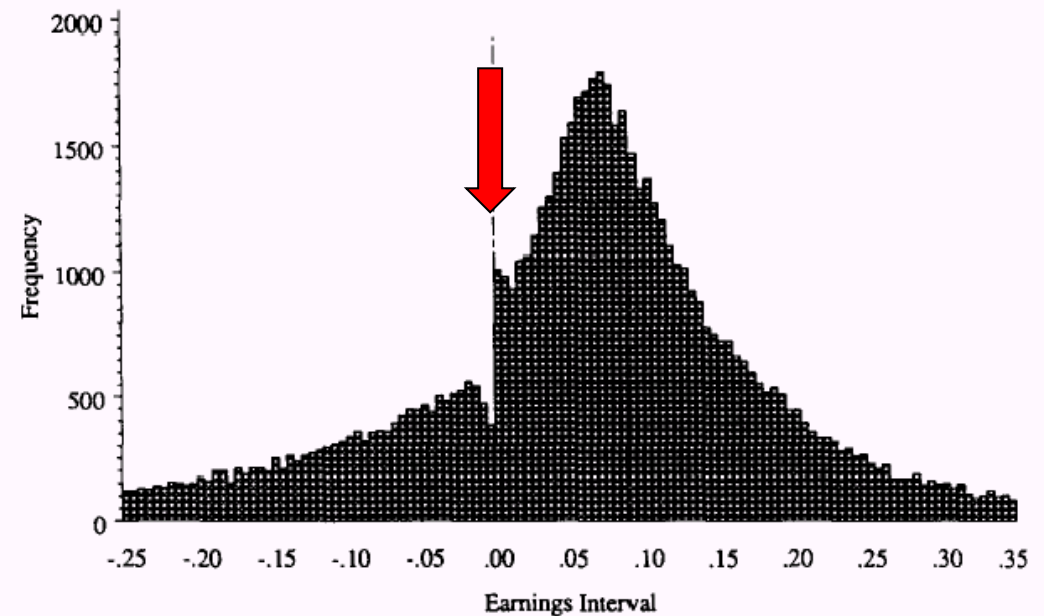
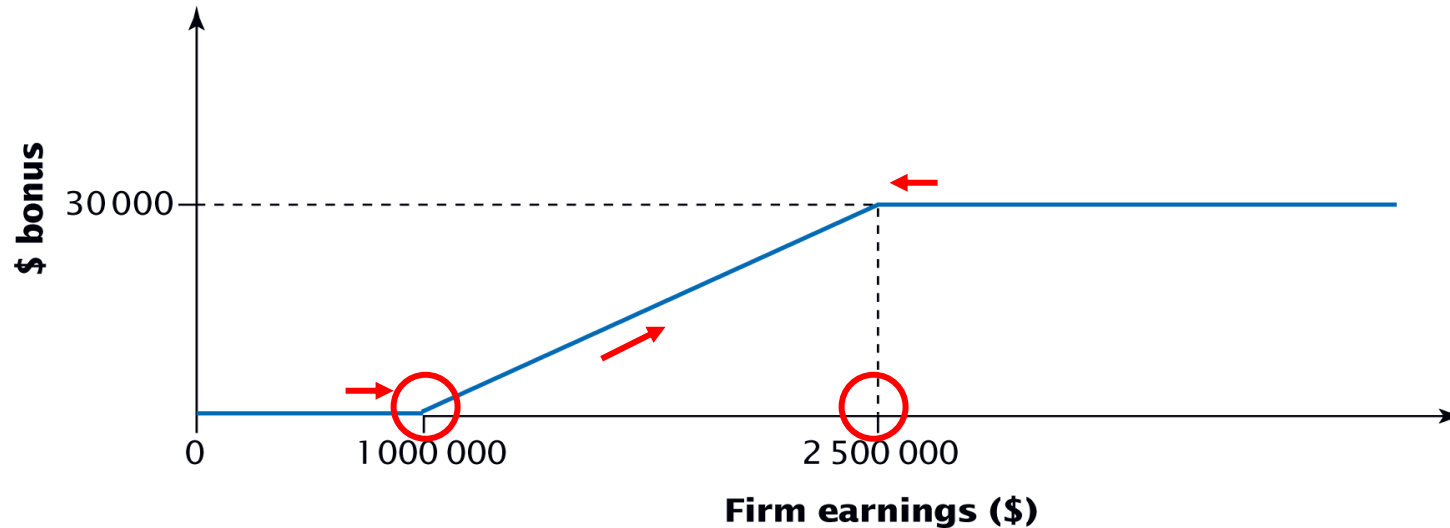


Fig. 3. The distribution of annual net income (Compustat item #172) scaled by beginning of the year market value (Compustat item #25 $\times$ Compustat item #199). The distribution interval widths are 0.005 and the location of zero on the horizontal axis is marked by the dashed line. When the interval width is 0.005, the first interval to the right of zero contains all observations in the interval [0.000, 0.005), the second interval contains [0.005, 0.010), and so on. 'Frequency' is the number of observations in a given earnings interval.

Examples of Contractual Incentives: Management Compensation

Bonus Plan Hypothesis

- To maximize cash bonus (current value + PV of future payment)
- Evidence: Healy [1985]
 - Typical bonus scheme with bogey and cap



**Figure 10.1: Allocation of funds to the bonus pool,
based on accounting profit (Healy, 1985)**

Detect EM: Discretionary accruals models

$$\text{Net Income} = \text{CFO} + \text{Total Accruals}$$

Discretionary
accruals/Unexpected
accruals/Abnormal
accruals

Proxy for Earnings
management

Non-Discretionary
accruals/Expected
accruals/Normal accruals

Considered as a function
of business operations.
E.g. a firm that mainly sells
on credit is expected to
have high accruals.



The ‘iron law’ of accrual-based earnings management

Net income = cash flows + accruals

Typical accruals:

- Depreciation and amortisation
- Increases/decreases in receivables/payables, inventories
- Prepayments/unearned revenue
- Provisions and write-offs

“Iron law” of earnings management:

accruals reverse

- In the long run, net income = cash flow
- $\sum \text{accruals} = 0$
- Slippery slope argument (how legitimate EM develops into fraud?)

More about accruals

- Typical accruals:
 - Depreciation and amortisation
 - Increases/decreases in receivables/payables, inventories
 - Prepayments/unearned revenue
 - Provisions and write-offs
- Aggressive vs Conservative Accounting:
 - Differ in the timing of revenue and expense recognition, with aggressive accounting aiming to maximize current period profits by shifting earnings from future to now.

Types of accruals (Qantas)

RECONCILIATION OF STATUTORY PROFIT FOR THE YEAR TO NET CASH FROM OPERATING ACTIVITIES

		2024	2023
	Notes	\$M	\$M
Statutory profit for the year		1,251	1,744
Adjusted for:			
Depreciation and amortisation	5	1,773	1,762
Impairment of assets and related costs	7	–	1
Hedging-related activities		47	(149)
Share of net (profit)/loss of investments accounted for under the equity method	14	(4)	44
Share-based payments expense	25	69	188
Net gain on disposal of assets	6	(18)	(4)
Impact of discount rate changes on provisions	7	3	(34)
Dividends received from investments accounted for under the equity method	14	5	12
Other items		59	43
Changes in other balance sheet items:			
Receivables		(166)	(13)
Inventories		(106)	(58)
Other assets		(176)	(162)
Payables		(89)	271
Revenue received in advance		206	741
Provisions		3	(28)
Deferred tax assets/liabilities and tax receivables/payables		584	727
Net cash inflow from operating activities		3,441	5,085

Prelude to an accounting quality analysis

- Understand the business
- Understand the accounting policies
- Understand the business areas where accounting quality is most doubtful
- Understand situations in which management are particularly **tempted** to 'manage earnings'

→ *incentives to report higher/lower profits?*

Accounting manipulation

- Situations where **manipulation** is likely :
 - **A weak governance structure:**
 - 1.inside management dominate the board;
 - 2.there is a weak **audit committee**
 - **Regulatory ratio requirements** (like capital ratios for banks and insurance companies) are likely to be violated
 - **Transactions with related parties** rather than at arm's length
 - **Special events** such as union negotiations and proxy fights
 - The firm is "in play" as a **takeover target**
 - Unexpected or (too) **frequent management changes**

Identifying accounting manipulation

- Identifying whether management is making choices that affect accounting is difficult.
 - How do we decide whether accruals truly reflect the underlying economic reality or not?
- Tend to focus on:
 - **Motive**: what incentives exist for management?
 - **Opportunity**: what conditions exist to allow for management choices?
 - **Means**: what accounting decisions are most useful and effective at managing in these circumstances?



Potential red flags

❑ We can always find signs when accounting choices have had effects:

- Unexplained changes in accounting when performance is poor
- Unexplained transactions that boost profits
- Abnormal increases in receivables relative to sales
- Abnormal increases in inventory relative to sales
- Increasing gap between accounting-based profit and cash flows

Modified auditor's reports are issued when the auditor believes the financial statements contain a material misstatement, or when the auditor is unable to obtain enough evidence to form an opinion.

- Increasing gap between accounting and tax profit
- Use of financing mechanisms
- Unexpected large asset write-offs (goodwill/ loan-other receivables)
- Large year end adjustments
- **Modified audit** opinions or change in auditors
- Poor internal governance systems
- Related-party transactions



Steps in Accounting Analysis

Step 1:
What is the objective?

- Is the objective to determine an independent valuation for a company?

Step 2:
What information is available?

- Do you have access to annual reports, interim reports, management reports and other reports?

Step 3:
Examine the audit report

- Who are the auditors?
- What is covered by the auditor report?

Step 4: look for unusual transactions and events

Unusual or non-recurring transactions and events can dominate the financial statements, including:

- Mergers and acquisitions
- Discontinued operations
- Major changes in management
- Potential lawsuits and major financing arrangements

The purpose is to consider if the major transactions make business sense or have been undertaken to achieve a certain accounting performance.

Step 5: assess accounting policies

An analyst needs to identify and evaluate a firm's key accounting policies, especially the methods used to measure critical factors and risks.

Accounting policy notes can be found along with relevant disclosures in the annual report. The explanations tend to be firm-specific and informative.

Another source by which to identify key accounting policies is the disclosure of critical accounting estimates in the annual report.



Key accounting policies

(B) CRITICAL ACCOUNTING ESTIMATES AND JUDGEMENTS

The preparation of the Consolidated Financial Statements requires Management to make judgements, estimates and assumptions that affect the application of accounting policies and reported amounts of assets, liabilities, income and expenses. It also requires the exercise of judgement in the process of applying the Group's accounting policies. The estimates and associated assumptions are based on historical experience and various other factors that are believed to be reasonable under the circumstances, the results of which form the basis for making the judgements about carrying values of assets and liabilities that are not readily apparent from other sources. Actual results may differ from these estimates.

Estimates and underlying assumptions are reviewed on an ongoing basis, as appropriate to the particular circumstances. Revisions to accounting estimates are recognised in the period in which the estimate is revised and in any future periods affected. In preparing this Report, areas of judgements made by Management in the application of Australian Accounting Standards that have a significant effect on the Consolidated Financial Statements and estimates with a significant risk of material adjustment in future periods are included in the following notes:

- Note 26(C)/Note 35(C) – Derivatives and Hedging Instruments
- Note 29 – Superannuation
- Note 33 – Contingent Liabilities and Legal Provisions
- Note 35(D) – Summary of Material Accounting Policies (Revenue Recognition)
- Note 22/Note 35(O) – Summary of Material Accounting Policies (Provisions).

Approaches to assess key accounting policies

Do the firm's accounting policies compare to industry norms?

Has the firm changed any of its policies?

Are there any prior period adjustments?

FIGURE 3.7 ASX consolidated statement of changes in equity

For the period ended	Note	Issued capital \$m	Retained earnings \$m	Restricted capital reserve \$m	Asset revaluation reserve \$m	Equity compensation reserve \$m	Total equity \$m
Opening balance at 1 July 2018		<u>3 027.2</u>	<u>666.7</u>	<u>71.5</u>	<u>168.4</u>	<u>11.7</u>	<u>3 945.5</u>
Change in accounting policies	10	–	(84.9)	–	0.6	–	(84.3)
Restated balance at 1 July 2018		<u>3 027.2</u>	<u>581.8</u>	<u>71.5</u>	<u>169.0</u>	<u>11.7</u>	<u>3 861.2</u>



Identify and assess key accounting policies

Improper Accounting Practice in the Report Pursuant to Section 704 of the Sarbanes-Oxley Act of 2002:

Improper Accounting Practice	Number of Enforcement Matters Involving Each Practice
Improper Revenue Recognition	126
Improper Expense Recognition	101
Improper Accounting in Connection with Business Combinations	23
Other Accounting and Reporting Issues:	130
Inadequate Disclosures in MD&A and Elsewhere	43
Failure to Disclose Related Party Transactions	23
Inappropriate Accounting for Non-monetary and Roundtrip Transactions	19
Improper Accounting for Foreign Payments in Violation of the FCPA	6
Improper Use of Off-balance Sheet Arrangements	3
Improper use of Non-GAAP Financial Measures	2

Source: SOX704report

Step 6: assess accrual quality

Financial statements are prepared using the accrual accounting.

Accrual accounting requires accountants to make estimates and judgements in allocating cash flows to past, current and future periods.

Accruals are the link between income and cash flows:

$$\text{Income} = \text{operating cash flows} + \text{accruals}$$

Use a reconciliation of net profit after tax with cash flow from operations to assess accrual quality.



Step 7: assess disclosure quality

Financial disclosures can make it more or less easy for an analyst to assess the firm's accounting performance.

While accounting rules require a certain amount of minimum disclosure, managers have considerable choice.

In assessing disclosure quality, an analyst could consider the following:

- Whether disclosures seem adequate
- Whether notes adequately explain a firm's current performance
- Whether segment disclosures seem adequate
- Whether disclosures are readable and understandable

Step 8: evaluate and conclude

The accounting analysis is a review of the information available, keeping in mind the objective of the analysis.

The accounting analysis should be a preliminary overview rather than an extensive investigation.

The accounting analysis will give the analyst an understanding of the available data.

The purpose of the accounting analysis is to assess the quality of the financial statements.



A Few More Points to Make

—

Accounting analysis pitfalls

- **Conservative** accounting is not necessarily 'good' accounting
- **Unusual accounting** is not necessarily questionable accounting
- Not all accounting policy and accruals changes are attributable to '*earnings management*' motives
- Modified **audit** opinions means auditors thinks there is something you need to pay attention to, not necessarily manipulation.



Real (earnings) management vs. accruals-based earnings management

Real (earnings) management (with direct cash flow effects):

- timing of marketing activities, R&D
- boosting sales (discount, providing generous terms to customers)
- selling a business branch
- timing of maintenance
- restructuring
- etc.

Accruals-based earnings management (through accounting choices, policies, estimates—no direct cash flow effects):

- bad debt allowances
- depreciation method
- adjusting the fiscal period
- provisions (warranty, future obligations, contingencies)
- etc.

Accruals reversal

Economic performance/reality: 10,000 bad debt will occur in Year 3

Situation 1:

Take bad debt allowances (when the transaction takes place in Year 0)

Dr. Bad debt expense (I/S in Year 0) 8,000

Cr. Bad debt allowances 8,000

In Year 3 write off bad debt

Dr. Bad debt expense (I/S in Year 3) 2,000

Dr. Bad debt allowances 8,000

Cr. Receivables 10,000

In Year 0, NI reduced by 8,000; In Year 3, NI reduced by 2,000



Accruals reversal

Economic performance/reality: 10,000 bad debt will occur in year 3

Situation 2:

No bad debt allowance consideration

That is, +8000 effect on NI in Year 0 (compared to Situation 1)

In Year 3 write off bad debt

Dr. Bad debt expense (I/S in year 3) 10,000

Cr. Receivables	10,000
-----------------	--------

One-time reduction on NI in Year 3 for 10,000

Compared to Situation 1, the overall effect on NI stays the same



Key Takeaways

- Accounting analysis is an essential step in analysing corporate financial reports.
- A methodology consisting of eight steps in analysing accounting data is presented.
- Sound accounting analysis improves the reliability of conclusions from financial analysis.



THE UNIVERSITY OF
MELBOURNE

See you at the seminar

